

MODULE-3

Relations, Functions and Algebraic Structures:

①

→ Set is a collection of objects

→ Repetition of elements not allowed

→ $a \in A$ a is an element of set A

→ N - natural no. $\{0, 1, 2, 3, \dots\}$

Z - integers $\{\dots, -3, -2, -1, 0, 1, 2, \dots\}$

Z^+ - positive integers $\{1, 2, 3, \dots\}$

R - set of all real numbers

R^+ - set of all positive real numbers

C - set of all complex numbers

Q - Set of all rational numbers

$$p/q, q \neq 0$$

→ $[a, b]$ $a \leq x \leq b$ (a, b) $a < x < b$

$[a, b)$ $a \leq x < b$ $(a, b]$ $a < x \leq b$

b not included

→ Sets can be elements of sets

$$\{\{1, 2, 3\}, a, \{b, c\}\}$$

$$\phi \neq \{\phi\}$$

→ Two sets are equal if and only if they have the same elements.

$$\text{eg: } \{1, 3, 5\} = \{3, 5, 1\}$$

$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

→ Subsets: The set A is a subset of B iff every element of A is also an element of B .

$$A \subseteq B$$

$\phi \subseteq S$ null set is a subset of every set

$S \subseteq S$ Set itself is a subset

→ Proper subsets ($A \subset B$)

$$A \subseteq B \text{ but } A \neq B$$

$$A = \{12, 24\}$$

$$B = \{12, 24, 36\}$$

$$A \subseteq B \text{ but } A \neq B$$

→ Cardinality

$|A|$: Cardinality of Set A

$$|\phi| = 0$$

↳ no. of distinct elements in A .

$$|\{1, 2, 3\}| = 3$$

→ Power Set ($P(A)$)

Set of all subsets of A

$$A = \{a, b\}$$

$$P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$$

$$\text{No. of subsets} = 2^n$$

$n \rightarrow$ no. of elements Cardinality.

↳ = no. of elements of power set (Cardinality)

→ Ordered pairs

$$(a, b) \neq (b, a)$$

→ Cartesian product

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

No. of elements of

$$A \times B = mn$$

$m \rightarrow$ (cardinality of A)

$n \rightarrow$ " of B .

Using

→ Set operations

(i) Union ($A \cup B$)

$$\{1, 2, 3\} \cup \{3, 4, 5\}$$

$$\{1, 2, 3, 4, 5\}$$



(ii) Intersection ($A \cap B$)

$$\{1, 2, 3\} \cap \{3, 4, 5\}$$

$$\{3\}$$

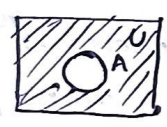
$$\{1, 2, 3\} \cap \{4, 5, 6\}$$

$$\phi$$

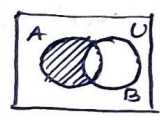


(iii) Complement (A')

$$A' = U - A$$



(iv) Difference ($A - B$)



$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{4, 5, 6, 7, 8\}$$

$$A - B = \{1, 2, 3\}$$

$$B - A = \{6, 7, 8\}$$

$$A - B \neq B - A$$

(v) Disjoint set

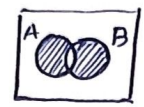
$$A \cap B = \phi$$



(vi) Symmetric Difference

$$A \oplus B$$

$$[(A - B) \cup (B - A)]$$



$$= [(A \cup B) - (A \cap B)]$$

• Set Identities

$$P \vee F \equiv F \quad P \wedge T \equiv P$$

$$\rightarrow A \cup \phi = A \quad A \cap U = A$$

Identity law

$$\rightarrow A \cup U = U \quad A \cap \phi = \phi$$

Dominant law

$$\rightarrow A \cup A = A \quad A \cap A = A$$

Idempotent law

$$\rightarrow A \cup B = B \cup A$$

Commutative

$$A \cap B = B \cap A$$

$$\rightarrow A \cup (B \cap C) = (A \cup B) \cap C$$

$$A \cap (B \cup C) = (A \cap B) \cup C$$

Associative

$$\rightarrow A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Distributive

$$\rightarrow \overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

De Morgan's law

$$\rightarrow A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Absorption law

$$\rightarrow A \cup \bar{A} = U \quad A \cap \bar{A} = \phi$$

Complement law

→ Proof of De-Morgan's Law

$$Q) A \cap B \subseteq \bar{A} \cup \bar{B}$$

$$am) x \in A \cap B \quad \text{assumption}$$

$$x \notin A \cap B \quad \text{def of complement}$$

$$\neg [(x \in A) \wedge (x \in B)] \quad \text{def of intersection}$$

$$\neg (x \in A) \vee \neg (x \in B) \quad \text{De-Morgan's law}$$

$$x \notin A \vee x \notin B \quad \text{def of negation}$$

$$x \in \bar{A} \vee x \in \bar{B} \quad \text{def of complement}$$

$$x \in \bar{A} \cup \bar{B} \quad \text{def of union}$$

$$Q) \bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$$

$$am) x \in \bar{A} \cup \bar{B}$$

$$(x \in \bar{A}) \vee (x \in \bar{B})$$

$$(x \notin A) \vee (x \notin B)$$

$$\neg (x \in A) \vee \neg (x \in B)$$

$$\neg [(x \in A) \wedge (x \in B)]$$

$$\neg (x \in A \cap B)$$

$$x \in \overline{A \cap B}$$

→ Using set-builder form

$$A \cap B =$$

$$\{x | x \in A \cap B\}$$

$$= \{x | \neg (x \in \overline{A \cap B})\}$$

$$= \{x | \neg (x \in A \cap B)\}$$

$$= \{x | \neg (x \in A) \vee \neg (x \in B)\}$$

$$= \{x | x \notin A \vee x \notin B\}$$

$$= \{x | x \in \bar{A} \vee x \in \bar{B}\}$$

$$= \{x | x \in \bar{A} \cup \bar{B}\}$$

$$= \bar{A} \cup \bar{B} //$$

(2)

Using Membership Table.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

A	B	C	$B \cap C$	$A \cup (B \cap C)$
1	1	1	1	1
1	1	0	0	1
1	0	1	0	1
1	0	0	0	1
0	1	1	1	1
0	1	0	0	0
0	0	1	0	0
0	0	0	0	0

$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	0	0
0	1	0
0	0	0

$$\begin{aligned}
 & x \in A \cup (B \cap C) \\
 & x \in A \vee x \in (B \cap C) \\
 & x \in A \vee [(x \in B) \wedge (x \in C)] \\
 & [x \in A \vee x \in B] \wedge [x \in A \vee x \in C] \\
 & [x \in A \cup B] \wedge [x \in A \cup C] \\
 & x \in (A \cup B) \cap (A \cup C) \\
 & (A \cup B) \cap (A \cup C) \\
 & \quad \quad \quad \underline{\quad \quad \quad}
 \end{aligned}$$

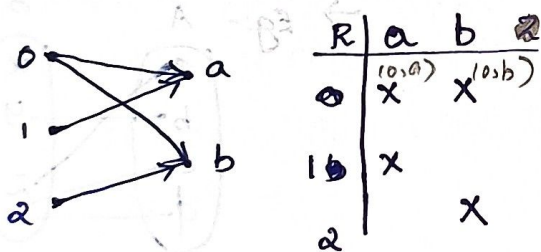
→ Binary relations

A Binary relation R from a set A to a set B is a subset of

$$R \subseteq A \times B$$

Eg:- $A = \{0, 1, 2\}$ $B = \{a, b\}$

$$R = \{(0, a), (0, b), (1, a), (2, b)\}$$



Eg:- let $B = \{1, 2, 3, 4\}$

$$R = \{(x, y) : x \text{ divides } y\}$$

ans:- $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 4), (2, 2), (3, 3), (4, 4)\}$

→ No. of relations = 2^{mn}

Domain → x values.

Range → y values

$$S = \{(1, 1), (2, 4), (3, 9)\}$$

$$\text{Domain}(S) = \{1, 2, 3\}$$

$$\text{Range} = \{1, 4, 9\}$$

→ Reflexive relation

Eg:- $A = \{1, 2, 3\}$

$$R_1 = \{(1, 1), (2, 2), (3, 3)\} \rightarrow \text{reflexive}$$

$$R_2 = \{(1, 1), (2, 2), (1, 2)\}$$

→ Not reflexive since

$$R_3 = \{(1, 2), (2, 3)\} \rightarrow \text{irreflexive. } (3, 3) \text{ is not there}$$

→ Symmetric relation

$$R_1 = \{(1, 2), (2, 1), (2, 2), (3, 4), (4, 3)\} \rightarrow \text{symmetric}$$

$$R_2 = \{(1, 1), (2, 2), (3, 3)\} \rightarrow \text{symmetric}$$

$$R_3 = \{(1, 1), (1, 2), (2, 1), (3, 4)\} \rightarrow \text{Not symmetric since } (4, 3) \text{ is not there}$$

$$R_4 = \{(1, 2), (4, 5)\} \rightarrow \text{Not symmetric}$$

→ Antisymmetric relation

Eg:- ~~{1,2,3}~~

$R_1 = \{(1,2), (2,1), (3,2)\} \rightarrow$ Not antisymmetric
if a, b & b, a exist then $a=b$ ($a \neq b$)

$R_2 = \{(0,1), (2,3), (4,5)\} \rightarrow$ Antisymmetric

↳ Nothing to compare.

→ Transitive relation

$R_1 = \{(1,2), (3,4), (5,6)\}$

is transitive nothing to compare

$R_2 = \{(1,2), (2,3)\} \rightarrow$ Not transitive
since $(1,3)$ not present.

→ Combining relation

Eg:- $A = \{1,2,3\}$

$B = \{1,2,3,4\}$

$R_1 = \{(1,1), (2,2), (3,3)\}$

$R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$

$R_1 \cup R_2 = \{(1,1), (2,2), (3,3), (1,2), (1,3), (1,4)\}$

$R_1 \cap R_2 = \{(1,1)\}$

$R_1 - R_2 = \{(2,2), (3,3)\}$

$R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$

→ If (x,y) members of R_1 and (y,z) members of R_2 then (x,z) is a member of $R_2 \circ R_1$.

→ Equivalence relation
binary R on a set A
A relation is an equivalence relation if it is reflexive, symmetric and transitive.

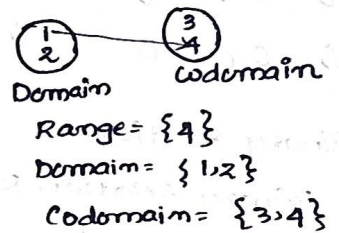
[Some more topic before fn]

• Functions ($f: A \rightarrow B$)

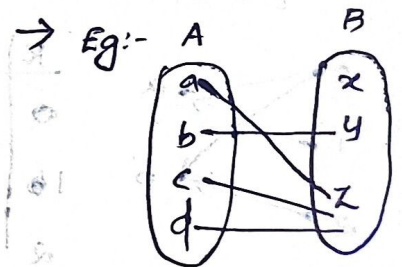
→ subset of $A \times B$ (a relation)

→ Domain - 1st set
Codomain - 2nd set

Range - set of all images of points of 1st set.



→ 2 functions are equal if they have
i) same domain,
ii) codomain, range.



vii)

$f\{a, b, c\}$ is $\{y, z\}$

viii) $f\{c, d\} = \{z\}$

i) $f(a) = x$

ii) image of $d = z$

iii) domain of $f = \{a, b, c, d\}$

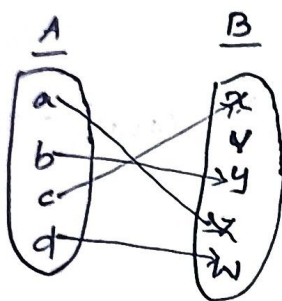
iv) codomain of $f = \{x, y, z\}$

v) Preimage of $y = b$

vi) $f(A) = \{y, z\}$

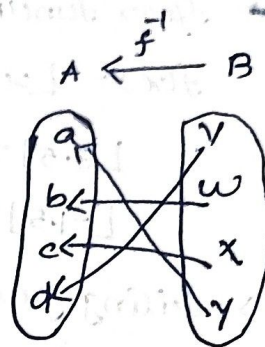
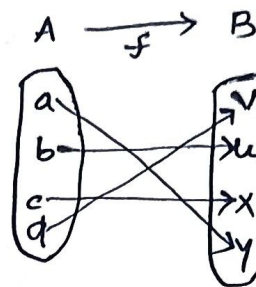
(3)

→ Injections [one-one] function



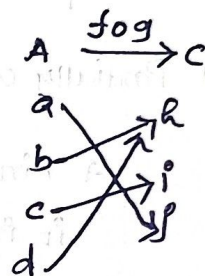
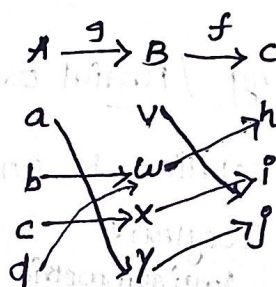
distinct elements
distinct image

distinct elements
should have
distinct image.



→ Composition

$$f \circ g(x) = f(g(x))$$



eg: $f(x) = x^2$ $g(x) = 2x + 1$

then $f \circ g = f(g(x))$

$$= f(2x + 1)$$

$$= \underline{\underline{(2x + 1)^2}}$$

$$g \circ f = g(f(x))$$

$$= g(x^2)$$

$$= \underline{\underline{2x^2 + 1}}$$

eg: $\{a, b, c\}$ and $\{1, 2, 3\}$

$$g(a) = b \quad f(a) = 3$$

$$g(b) = c \quad f(b) = 2$$

$$g(c) = a \quad f(c) = 1$$

an: $f \circ g(a) = f(g(a))$

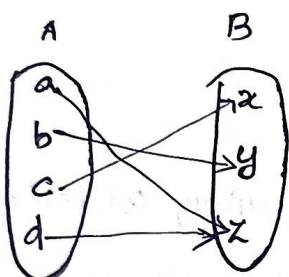
$$= f(b) = \underline{\underline{2}}$$

$$f \circ g(b) = f(g(b))$$

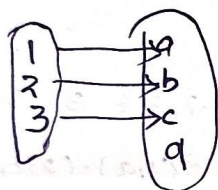
$$= f(c) = \underline{\underline{1}}$$

$$f \circ g(c) = f(g(c)) = f(a) = \underline{\underline{3}}$$

→ Surjections [onto] function



must need
a preimage.



not an onto

→ Bijections

Function which is both one-one & onto

→ $f(x) = x^2$ from a set of integers
to the set of integers onto?

an: No, because there is no integer x
with $x^2 = -1$.

→ Inverse function

Let f be a bijection from $A \rightarrow B$

then f^{-1} is a function from $B \rightarrow A$

→ No inverse exist unless f is a
bijection.

→ Floor function

$$f(x) = \lfloor x \rfloor$$

$$\lfloor 3.5 \rfloor = 3$$

$$\lfloor -1.5 \rfloor = -2$$

→ Ceiling function

$$f(x) = \lceil x \rceil$$

$$\lceil 3.5 \rceil = 4$$

$$\lceil -1.5 \rceil = -1$$

→ Identity function

$$\text{Let } A = \{1, 2, 3, 4\}$$

Identity fn of A

$$f = \{(1,1), (2,2), (3,3), (4,4)\} = I_A$$

•) Partially ordering / Partial order relation

A binary relation R on a set A is said to be partially ordering if

it is → Reflexive
Antisymmetric
Transitive

•) Representation of relation

↳ Relation matrix

Digraph / Directed graph.

Eg:- $A = \{1, 2, 3, 4\}$ R is a relation on A.

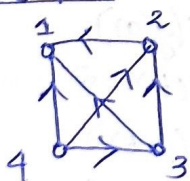
$$R = \{(1,2), (1,3), (2,4), (3,2)\}$$

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Eg: Let $X = \{1, 2, 3, 4\}$ $R = \{(x,y) | x > y\}$

$$\text{an) } R = \{(2,1), (3,1), (4,1), (3,2), (4,2), (4,3)\}$$

digraph.



Matrix.

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$\text{Indegree}(1) = 3$$

$$\text{outdegree}(1) = 0$$

$$\text{Indegree}(2) = 2$$

$$\text{outdegree}(2) = 1$$

$$\text{Indegree}(3) = 1$$

$$\text{outdegree}(3) = 2$$

$$\text{Indegree}(4) = 0$$

$$\text{outdegree}(4) = 2$$

Digraph.

$$x R y$$

$$x R x$$

$$x R y \text{ \& } y R x$$

$$R = \{(x,y), (y,z), (z,x)\}$$

$$R = \{(x,y), (y,y)\}$$

(4)

Composition of relation. R is a relation from $A \rightarrow B$ S is a relation from $B \rightarrow C$ Then $R \circ S$ is a relation from $A \rightarrow C$ If (x, y) member of R , (y, z) member of R_2 Then (x, z) member of $R \circ R_1$

~~$$\text{eg: } A = \{1, 2, 3\} \quad B = \{2, 3, 4, 8, 12\} \quad C = \{13, 14, 22\}$$~~

~~$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 6), (2, 8), (2, 12)\}$$~~

~~$$S = \{(2, 13), (2, 14), (3, 13), (3, 22), (4, 22)\}$$~~

$$\text{eg:- } R = \{(1, 2), (3, 4), (2, 2)\} \quad S = \{(4, 2), (2, 3), (3, 1), (1, 3)\}$$

$$\text{Find a) } R \circ S = \{(1, 3), (3, 2), (2, 3)\}$$

$$\text{b) } S \circ R = \{(4, 2), (2, 4), (3, 2), (1, 4)\}$$

$$\text{c) } R \circ (S \circ R) = \{(1, 4), (3, 2), (2, 4)\}$$

$$\text{d) } (R \circ S) \circ R = \{(1, 4), (3, 2), (2, 4)\}$$

$$\text{e) } S \circ S = \{(4, 3), (2, 1), (3, 3), (1, 1)\}$$

$$\text{f) } R \circ R = \{(1, 2), (2, 2)\}$$

$$\text{g) } R \circ R \circ R = \{(1, 2), (2, 2)\}$$

Composition of function.Let $x = \{1, 2, 3\}$ and f, g, h be functions from x to x given by

$$f = \{(1, 2), (2, 3), (3, 1)\}$$

$$g = \{(1, 2), (2, 1), (3, 3)\}$$

$$h = \{(1, 1), (2, 2), (3, 1)\}$$

an) fog

$$f \circ g(1) = f(g(1)) = f(2) = 3 = \{(1, 3), (2, 2), (3, 1)\}$$

$$f \circ g(2) = f(g(2)) = f(1) = 2$$

$$f \circ g(3) = f(g(3)) = f(3) = 1$$

goh

$$g \circ h(1) = g(h(1)) = g(1) = 2$$

$$g \circ h(2) = g(h(2)) = g(2) = 1 = \{(1, 2), (2, 1), (3, 2)\}$$

$$g \circ h(3) = g(h(3)) = g(1) = 2$$

→ Partition of a set

If S is a non-empty set, a collection of disjoint non-empty ~~sets~~ subsets of ' S ' whose union is ' S ' is called a partition of ' S '.

Eg:- $A = \{7, 8, 9\}$

Partitions of A are:

1: $\{\{7\}, \{8\}, \{9\}\}$

2: $\{\{7, 8\}, \{9\}\}$

3: $\{\{7\}, \{8, 9\}\}$

4: $\{\{7, 9\}, \{8\}\}$

5: $\{\{7, 8, 9\}\}$

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Determine whether the following is a partition of A or not.

(i) $P_1 = \{\{1, 2, 3, 4\}, \{1, 3, 5, 6, 7, 8\}\}$ $\times$ 1, 3 belongs to 2 subsets.
are not disjoint subsets.

$P_2 = \{\{1, 2, 3, 4\}, \{6, 8\}\}$ $\times$ 5, 7 is not included
doesn't form union to S

→ Equivalence Class

Let R be an equivalence relation on a set A . For each $x \in A$, the equivalence class of x , denoted as $[x]$ such that

$$[x] = \{y \in A \mid y R x\}$$

Eg: $A = \{1, 2, 3\}$

$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

$[1]_R = \{1, 2\}$

$[2]_R = \{1, 2\}$

$[3]_R = \{3\}$

→ Quotient set of A

Collection of all equivalence classes

$A/R = \{[a] \mid a \in A\}$

no need of repetition in set

→ Partial Order Set (Poset)

Set A together with a partially ordering relation R on the set A is called a POSET.

Denoted by (A, R) .

Eg: $A = \{1, 2, 3, 4\}$ R -divides

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$

reflexive ✓
antisymmetric ✓
transitive ✓
Partial ordering
So POSET
(A, R)

(5)

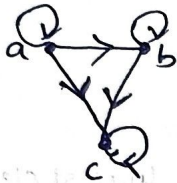
Hasse diagram.

*) The poset can be represented by using a diagram known as Hasse diagram.

*) It can be obtained from its digraph by

- removing all loops
- removing all edges
- removing arrows.

*) Eg:-

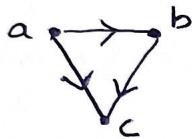


reflexive
antisymmetric
+ transitive

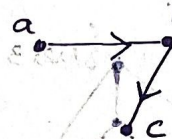
(A, R) is a poset

So we can construct Hasse diagram.

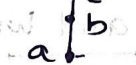
① removing loops



② removing all edges



③ removing arrows

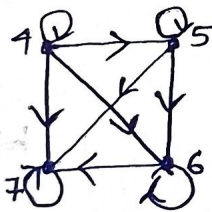


Hasse diagram.

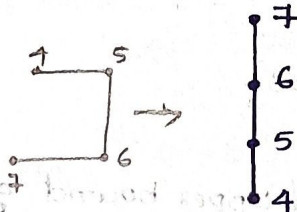
Eg:- Set $A = \{4, 5, 6, 7\}$ R be the relation $\leq$ on A .

an) $R = \{(4, 4), (4, 5), (4, 6), (4, 7), (5, 5), (5, 6), (5, 7), (6, 6), (6, 7), (7, 7)\}$

Digraph.



Hasse diagram.



Note:- Divides (1)

$x = \{2, 3, 6, 12, 24, 36\}$



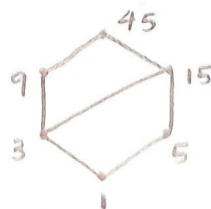
Eg:- $A =$ Set of all factors of 45

relation (1)

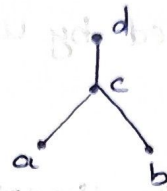
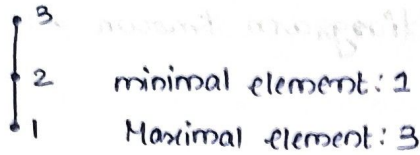
or $(D_{45}, 1)$ divides relation.

$A = \{1, 3, 5, 9, 15, 45\}$

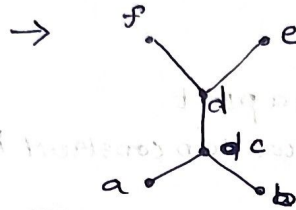
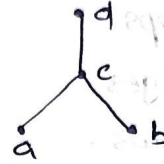
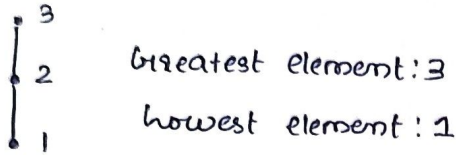
$I = \{(1, 1), (1, 3), (1, 5), (1, 9), (1, 15), (1, 45), (3, 3), (3, 9), (3, 15), (3, 45), (5, 5), (5, 15), (5, 45), (9, 9), (9, 45), (15, 15), (15, 45), (45, 45)\}$



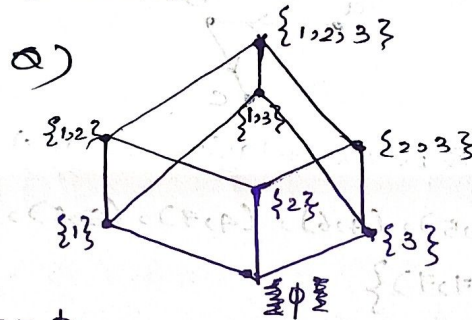
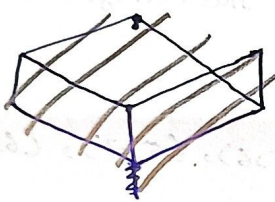
→ Minimal and maximal element



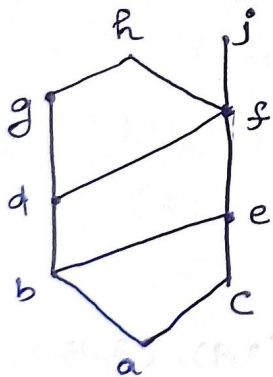
→ Greatest and lowest element



→ Upper bound and lower bound



$P(U) \rightarrow$ power set of U
 $A \subseteq \rightarrow$ subset of A
cube lattice diagram



lower bound & upper bound of

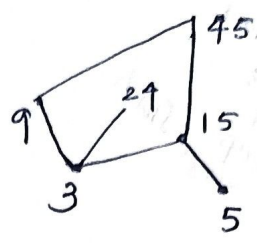
a) $\{a, b, c\}$ - LB - $\{a\}$
UP - $\{e, f, h, i\}$

$\{j, h\}$ - LB - $\{f, d, b, a, c, e\}$
UP - nil.

$\{a, c, d, f\}$ - LB - $\{a\}$
UP - $\{h, i\}$

LUB $\Rightarrow$ lowest upper bound or Supremum

LUB
lowest upper bound / supremum
GLB
Greatest lower bound / infimum.

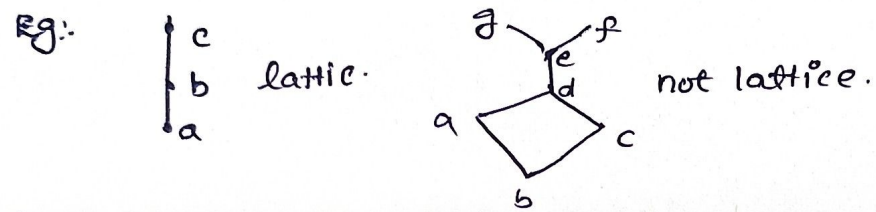


Maximal element: 24, 45
minimal element: 3, 5
lowest & greatest element = no

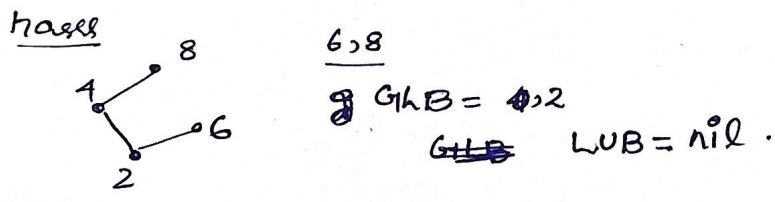
upper bound $\{3, 5\} = \{15, 45\}$
lower bound $\{15, 45\} = \{3, 5, 15\}$
 $LUB \{3, 5\} = \underline{15}$ } not value.
 $GLB \{15, 45\} = \underline{15}$

→ lattice. $(L, \wedge, \vee)$

It is a poset in which every pair of elements have both GLB & LUB.



Eg: POSSET $(\{2, 4, 6, 8\}, |)$
relation = $\{(2, 4), (2, 6), (2, 8), (4, 8), (2, 2), (4, 4), (6, 6), (8, 8)\}$



Note:

$MR_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $MR_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
a) $R_1 \cup R_2 \rightarrow \oplus$ add. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
b) $R_1 \cap R_2 \rightarrow \times$ multiply. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$
c) $R_1 \circ R_2$
d) $R_2 \circ R_1$ $(MR_2 \times MR_1)$ $\rightarrow$ matrix multiplication

e) $R^{-1} \rightarrow$ transpose of matrix.
f) $R^2 \rightarrow R \times R \rightarrow$ matrix multiplication
g) $\bar{R} \rightarrow$ switch 0's & 1's

an element of B.

$A \subseteq B$ $\emptyset \subseteq S$ null set is a subset of every set
 $S \subseteq S$ Set itself is a subset

No. of elements of $A \times B = mn$
 $m \rightarrow$ cardinality of A
 $n \rightarrow$ " of B